Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2025	ENGI3311-DE01

Title: (BEng/MEng)

Geotechnics 3

Time Allowed:	2 hours
Additional Material provided:	No
Reporting problems during the exam:	Queries about the instructions for this exam (rubric) should be directed to: eng.boe@durham.ac.uk. If you experience any difficulties uploading your exam answers to Gradescope report these immediately to: eng.boe@durham.ac.uk.

Instructions to Candidates:	This is a synchronous, time restricted exam. All submissions must be uploaded to Gradescope as per the published timetable. Answer ALL questions. Please answer each question on a new page. All workings must be shown. This is an online open-book exam. • *The examination attempt must be all your own work. You must not interact with another person (either one-to-one, by telephone, on the internet, in chat rooms, on social media, via smart watches, etc.), or use cheat sites, etc. whilst completing the exam. You must not discuss any examination papers with friends, relatives, current or past students of this or any other University before the end of the exam submission window.* • *Candidates are expected to answer all questions in their own words. Should close paraphrasing, direct quotation be necessary candidates must acknowledge the source of any material used in their answers (this includes text, images, diagrams, charts, tables and/or graphs). Full referencing is not expected, but a brief indication of the source of the material (e.g. name of author) must be given.* • *In line with the Department of Engineering document entitled "Student Guidance on the Use of Generative Artificial Intelligence and Related Technology for the Department of Engineering" policy in assessments, the use of generative artificial intelligence tools and related technologies is strictly prohibited for this exam.*

Revision:	

Q1

A cylindrical storage tank is to be constructed at a site close to a river. The responsible geotechnical engineer wants to investigate the ground conditions of the clay deposit where the tank is going to be built (see Figure Q1a). The saturated unit weight of the clay is 16 kN/m^3 and its friction angle is 23°.

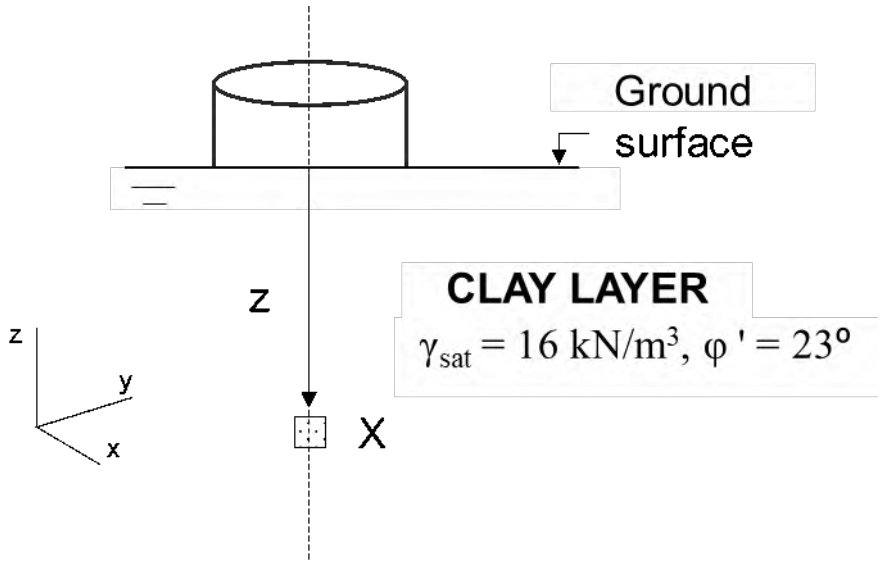


Figure Q1a (Not to scale)

(a) Assuming that the soil is normally consolidated, find expressions for the variation of the effective vertical and horizontal stresses with depth, z. Sketch your results plotting stress against depth. Assume that the water table is at the ground surface. [15%]

(b) Find also the expressions for the variation of the effective vertical and horizontal preconsolidation pressures with depth. Sketch your results plotting stress against depth. [15%]

(c) If the water table is lowered to 1m from the ground surface, express how the effective vertical and horizontal stresses vary with depth in this new situation. Assume that the capillary rise in this clay is larger than 1 m. Sketch also your stress profile results. [20%]

(d) A period of heavy rainfall brings back the water table to the ground surface. Is the clay at this new scenario normally consolidated or overconsolidated? Find an expression for the variation of OCR with depth and use it to express the variation of effective horizontal stress with depth. [20%]

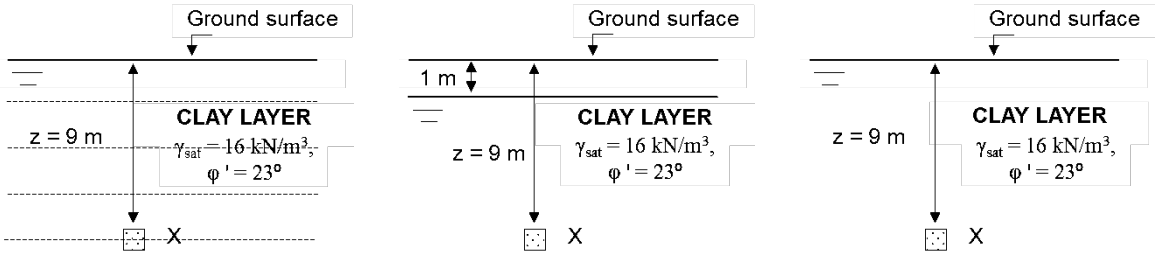
(e) Consider an element of soil X at a depth of 9 m, as indicated in Figure Q1b. Consider also that the previous loading history of this element of soil has only involved one dimensional vertical straining. Assuming that the stress state of this element of soil is axisymmetric (i.e., $p' = (\sigma'_v + 2\sigma'_h)/3$ and $q = (\sigma_v - \sigma_h)$), plot the stress path in the q:p' plane of this soil element including the following three stages:

- Stage A - corresponding to the soil formation during deposition in horizontal layers as part of the process of sedimentation,
- Stage B - corresponding to the stress state after the decrease of water table described in Question Q1(c) and

continued

- Stage C - corresponding to the stress state after the increase of water table described in Question Q1(d).

[30%]



Stage A Stage B Stage C

Figure Q1b (Not to scale)

continued

Q2

A layer of saturated clay with a thickness of 8 m rests on an impermeable rock formation. The clay has a saturated unit weight of 18 kN/m^3, an undrained shear strength of 100 kPa, an effective internal friction angle of 30° and an effective cohesion of 10 kPa. A geotechnical engineer wants to design a square concrete shallow foundation of width 1.5 m subjected to a vertical load V and a horizontal load H. Consider that the footing is placed 1m below the ground surface (see Figure Q2a). Take account of the weight of the footing in your calculations by assuming that the unit weight of the concrete is 25 kN/m^3. The water table is at the ground surface.

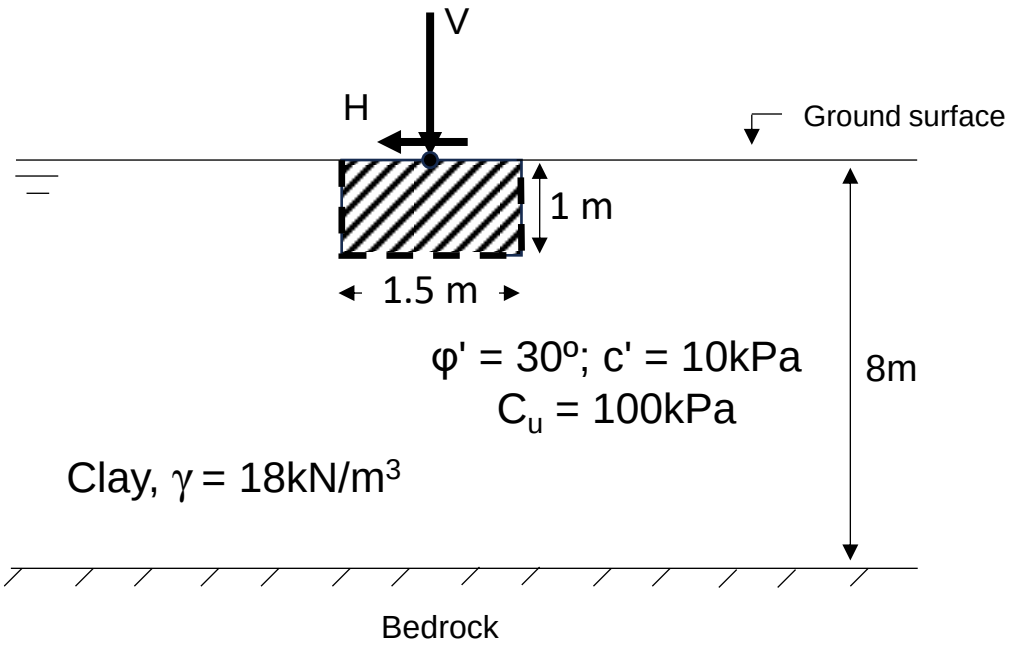


Figure Q2a (Not to scale)

(a) Assuming that H = 0kN and V = 1,000 kN, calculate the factor of safety of the foundation against bearing capacity failure in the short term. Justify all the steps followed in your calculations. [30%]

(b) Assuming that H = 0kN and V = 1,000 kN, calculate the factor of safety of the foundation against bearing capacity failure in the long term. Justify all the steps followed in your calculations and discuss your result when compared against that obtained in (a). [30%]

Assume now that H = 50 kN and V remains as 1,000kN.

(c) Calculate the factor of safety of the foundation against bearing capacity failure in the short term. Discuss your result when compared against the factor of safety values obtained in (a) and (b). [40%]

continued

Q3

A concrete retaining structure has been constructed in York to prevent flooding from the river Ouse (see Figure Q3). The water table level behind the wall is at 2 m below the ground surface (with no capillary rise) and the height of the river in front of the wall is of 2 m. The soil behind the wall comprises a 5 m thick layer of sandy soil, overlying a layer of clay. The total height of the wall is 9 m with a base of 5 m, as indicated in Figure Q3. The unit weight of the concrete is 25 kN/m^3. Assume hydrostatic conditions.

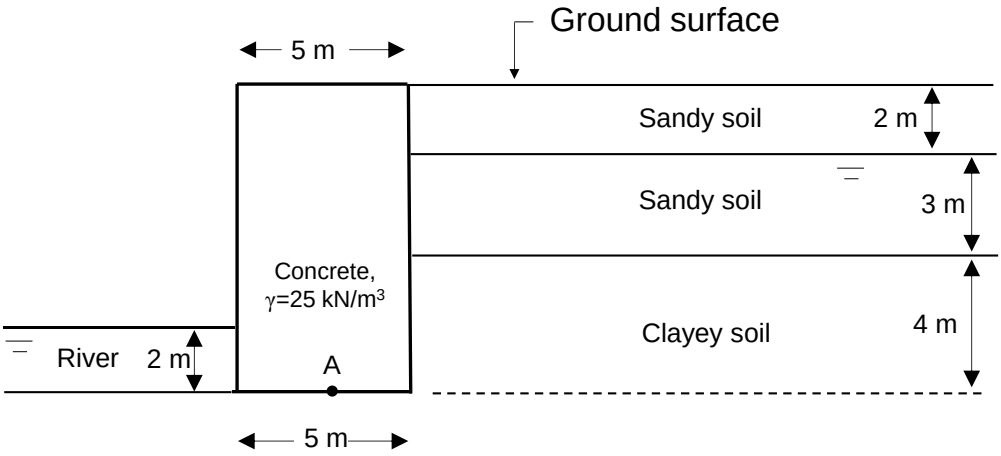


Figure Q3 (Not to scale)

The geotechnical information of the ground behind the wall is as follows:

Sandy soil:

$\gamma_{dry} = 18\ kN/m^3$, $\gamma_{sat} = 20\ kN/m^3$, $\varphi' = 33°$, $c' = 0$ and $\varphi_w{}' = \varphi'$ (friction angle wall -sand)

Clayey soil:

$\gamma_{sat} = 17\ kN/m^3$, $\varphi' = 15°$ and $c' = 20\ kPa$. Assume that when $\varphi_w{}' = \varphi'$ (friction angle wall – clay) and $c_w{}' = 0.5c'$ (cohesion wall-clay), the active coefficients of earth pressures for the clay can be assumed as $K_{a1} = 0.5176$ and $K_{a2} = 1.6047$.

(a) Using Coulomb's theory of lateral earth pressure, calculate the resultant horizontal forces H acting on the wall. Sketch all the horizontal forces acting in the wall. [50%]

(b) Using Coulomb's theory of lateral earth pressure, calculate the resultant vertical forces acting on the wall (Point A in Figure Q3). Sketch all vertical forces acting in the wall. [50%]

continued

Phase relationships

$e = \frac{V_v}{V_s}$	$S_r = \frac{V_w}{V_v}$	$v = 1 + e$	$n = \frac{e}{1+e}$	$w = \frac{M_w}{M_s}$
$\rho_s = G_s \rho_w$	$wG_s = S_r e$	$\rho = \frac{M_s + M_w}{V_s + V_v}$	$\rho = \frac{(G_s + S_r e)\rho_w}{1+e}$	$\gamma = \rho g$
$\gamma = \frac{(G_s + S_r e)\gamma_w}{1+e}$	$\gamma' = \gamma_{sat} - \gamma_w$	$\rho_d = \frac{G_s \rho_w}{1+e}$	$\rho_d = \frac{\rho}{1+w}$	$\gamma_w \approx 10$ kN/m^3

Stresses and pore pressures in soils:

$\sigma_v = \rho g z = \gamma z$	$u_w = \rho_w g z_w = \gamma_w z_w$	$\sigma_v' = \sigma_v - u$	$u_a - u_w = \frac{2T}{R}$	$h_c = \frac{2T}{\gamma_w R}$

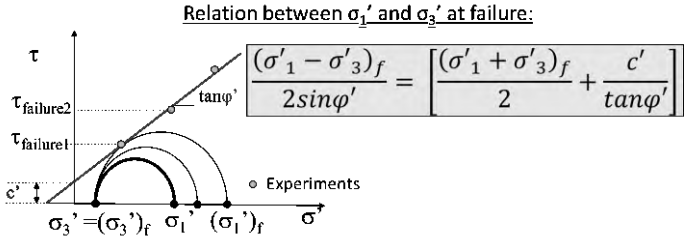
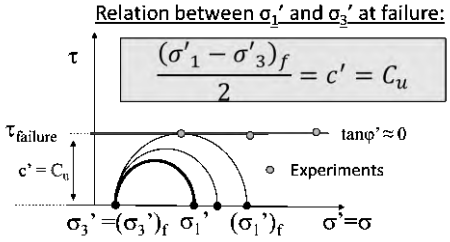
Elastic stress-strain relations:

The volumetric strain is: $\Delta\varepsilon_v = \Delta\varepsilon_1 + \Delta\varepsilon_2 + \Delta\varepsilon_3 = \Delta\varepsilon_x + \Delta\varepsilon_y + \Delta\varepsilon_z$

$\Delta\varepsilon_x = \frac{1}{E'}(\Delta\sigma_x' - \nu'\Delta\sigma_y' - \nu'\Delta\sigma_z')$	$\Delta\gamma_{xy} = \frac{\Delta\tau_{xy}}{G'}$	--
$\Delta\varepsilon_y = \frac{1}{E'}(\Delta\sigma_y' - \nu'\Delta\sigma_x' - \nu'\Delta\sigma_z')$	$\Delta\gamma_{yz} = \frac{\Delta\tau_{yz}}{G'}$	$K' = \frac{E'}{3(1-2\nu')}$
$\Delta\varepsilon_z = \frac{1}{E'}(\Delta\sigma_z' - \nu'\Delta\sigma_y' - \nu'\Delta\sigma_x')$	$\Delta\gamma_{zx} = \frac{\Delta\tau_{zx}}{G'}$	$G' = \frac{E'}{2(1+\nu')}$
$\Delta\varepsilon_v = \frac{\Delta p'}{K'}$		$\Delta\varepsilon_q^e = \frac{\Delta q}{3G'}$
$p' = \frac{\sigma'_1 + \sigma'_2 + \sigma'_3}{3}$		$q = \left[\frac{(\sigma_1' - \sigma_2')^2 + (\sigma_2' - \sigma_3')^2 + (\sigma_3' - \sigma_1')^2}{2}\right]^{1/2}$

Where σ'_1, σ'_2 and σ'_3 are the principal effective stresses (i.e. $\sigma'_1 \geq \sigma'_2 \geq \sigma'_3$)

Failure envelopes:

$\tau = c' + \sigma' \tan\varphi'$ (Mohr-Coulomb)	$\tau = C_u$ (Tresca)
Relation between σ_1' and σ_3' at failure: $\frac{(\sigma'_1 - \sigma'_3)_f}{2\sin\varphi'} = \left[\frac{(\sigma'_1 + \sigma'_3)_f}{2} + \frac{c'}{\tan\varphi'}\right]$ 	Relation between σ_1' and σ_3' at failure: $\frac{(\sigma'_1 - \sigma'_3)_f}{2} = c' = C_u$ 

Ground water flow:

$h = z + \frac{u}{\rho g} = z + \frac{u}{\gamma_w}$	$q_x = Aki_x = Ak\left(-\frac{\partial h}{\partial x}\right)$	$v_x = ki_x$	$v_{true,x} = v_x \frac{1}{n}$

continued

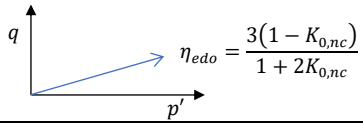
Constant head permeameter:	Falling head:	Pumping test:
$q = Ak\dfrac{h_1 - h_2}{L}$	$ln\left(\dfrac{h_0}{h}\right) = \dfrac{Ak}{aL}t$	$\left(H^2 - h^2\right) = \dfrac{q}{\pi k}(ln\, R_0 - ln\, r)$

Laplace's Equation (3D)	Laplace's Equation (2D, flownets)	
$\dfrac{\partial^2 h}{\partial x^2} + \dfrac{\partial^2 h}{\partial y^2} + \dfrac{\partial^2 h}{\partial z^2} = 0$	$\delta h = \dfrac{\Delta h}{N_d}$	$q = k\dfrac{N_f}{N_d}\Delta h;\ u = \gamma_w(h - z)$

Critical hydraulic gradient:	$i_c = \dfrac{\gamma - \gamma_w}{\gamma_w}$

1D flow through an unconfined aquifer:	1D flow through a confined aquifer:
$q_x = \dfrac{k(H_1^2 - H_2^2)}{2L}$	$q_x = Aki_x = -Ak\dfrac{\partial h}{\partial x} = -k\dfrac{H_2 - H_1}{L}D$
$h = \sqrt{\dfrac{(H_2^2 - H_1^2)}{L}x + H_1^2}$	$h = \dfrac{H_2 - H_1}{L}x + H_1$

1D straining (oedometric test conditions):



1D normal compression line:	1D swelling line:
$e = e_1 - C_c\, log_{10}\, \sigma'_v$	$e = e_2 - C_s\, log_{10}\, \sigma'_v$

$OCR_{1D} = \dfrac{\text{max. past value of } \sigma'_v}{\text{current value of } \sigma'_v} = \dfrac{\sigma'_{vc}}{\sigma'_v}$	$\sigma'_h = K_0\sigma'_v$	$K_{0nc} \approx 1 - sin\,\varphi'$	$K_0 \approx (1 - sin\,\varphi')\sqrt{OCR}$

Settlement of a very wide foundation:		
$\dfrac{\Delta H}{H_0} = \dfrac{e_0 - e_1}{1 + e_0}$	$\Delta H = \dfrac{H_0}{1 + e_0}\left[C_s\, log_{10}\dfrac{\sigma'_{vc}}{\sigma'_{v0}} + C_c\, log_{10}\dfrac{\sigma'_{v1}}{\sigma'_{vc}}\right]$	$\Delta\varepsilon_v = m_v\Delta\sigma'_v$

1D consolidation:		
$\dfrac{\partial u_e}{\partial t} = c_v\left(\dfrac{\partial^2 u_e}{\partial z^2}\right)$	$c_v = \dfrac{k}{m_v\gamma_w}$	$T_v = \dfrac{c_v t}{d^2}$

Axisymmetric conditions (triaxial test): σ_a' and σ_r' are the principal effective axial and radial stresses

$\Delta\varepsilon_v = \Delta\varepsilon_a + 2\Delta\varepsilon_r$	$\Delta\varepsilon_q = \dfrac{2}{3}(\Delta\varepsilon_a - \Delta\varepsilon_r)$
$p' = \dfrac{1}{3}(\sigma_a' + 2\sigma_r')$	$q = \sigma_a - \sigma_r$

continued

Iso-normal compression line:	Iso-swelling line:
$v = N - \lambda \ln p'$	$v = v_\kappa - \kappa \ln p'$

Critical states:	
$q = Mp'$	$v = \Gamma - \lambda \ln p'$

where Γ=N-(λ-κ)ln(2);

Modified Cam Clay yield curve *f*:		$q^2 - M^2 p'(p'_0 - p') = 0$	
Plastic volumetric strains	$\Delta\varepsilon_v^p = \frac{(\lambda-\kappa)}{v} ln\left(\frac{p'_{0B}}{p'_{0A}}\right)$	Plastic deviatoric strains	$\Delta\varepsilon_q^p = \frac{2\eta}{M^2-\eta^2}\Delta\varepsilon_v^p$
Stress ratio		$\eta = \frac{q}{p'}$	
Friction angle and critical state gradient relation:		$M = \underbrace{\frac{q}{p'}}_{at\ critical\ state} = \frac{6\sin\varphi'}{3-\sin\varphi'}$	

Overconsolidation ratio (OCR in terms of p')	$OCR = \frac{max.\ past\ value\ of\ p'}{current\ value\ of\ p'} = \frac{p'_0}{p'}$

Partial factors Eurocode 7 & factor of safety: (Bond & Harris, 2008; *Decoding EC7*)

$q_{Ed} \le q_{Rd}$ where, q_{Ed} is the design pressure and q_{Rd} is the design resistance	CASE	ACTIONS			Soil properties	
		Permanent		Variable	tan φ', c'	C_u
		Unfavourable	Favourable	Unfavourable		
	C	1.00	1.00	1.30	1.25	1.4

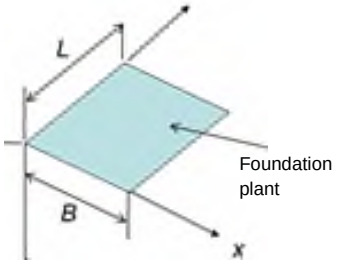
Factor of safety, FoS:	$FoS = \frac{available\ shear\ force,\ T_f}{allowable\ working\ load,\ T}$; $FoS = \frac{available\ bearing\ capcity,\ \tau_f}{allowable\ bearing\ capacity,\ \tau}$

Shallow foundation settlement, δ:

Cohesionless soils:

Burland & Burbage:

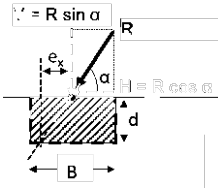
$$\delta\ (mm) = f_s \cdot f_l \cdot f_t \cdot \left[\left(q'_n - \frac{2}{3}p'_0\right) \cdot B^{0.7} \cdot I_c\right]$$



continued

$f_s = \left(\frac{1.25 \cdot L/B}{\frac{L}{B} + 0.25}\right)^2$	$f_l = \frac{H}{z_l}\left(2 - \frac{H}{z_l}\right)$	$f_t = 1 + R_3 + R \cdot log\left(\frac{t}{3}\right)$
Cohesive soils:	$\delta = \delta_i + \delta_c$ where,	
$\delta_i\ (mm) = \mu_1 \cdot \mu_0 \cdot q_n \frac{B}{E_u}$	$\delta_c\ (mm) = \mu_g \cdot \delta_{oed}$ where, $\delta_{oed}\ (mm) = m_v \cdot \sigma'_v \cdot H$	

Bearing capacity: depth factors are not permitted within EC7; base and ground factors ignored

Brinch-Hansen (drained all soils)		
$q_f = c' \cdot N_c \cdot s_c + p'_{ov} \cdot N_q \cdot s_q + \frac{1}{2} \cdot \gamma' \cdot B' \cdot N_\gamma \cdot s_\gamma$ where,		
$N_q = e^{\pi \tan \varphi'} \cdot \left(\tan(45 + \varphi'/2)\right)^2$	$s_q = 1 + (B'/L') \cdot \sin \varphi'$ (rectangular); $s_q = 1 + \sin \varphi'$ (circular & square)	
$N_c = \left(N_q - 1\right) \cdot \cot \varphi'$	$s_c = \left(s_q \cdot N_q - 1\right)/\left(N_q - 1\right)$ (all)	
$N_\gamma = 2 \cdot \left(N_q - 1\right) \cdot \tan \varphi'$	$s_\gamma = 1 - 0.3 \cdot (B'/L')$ (rectangular; $s_\gamma = 0.7$ (circular and square)	
$i'_q = (1 - 0.7tan\delta')^3$	$i'_c = \frac{i'_q N_q - 1}{N_q - 1}$	$i'_\gamma = (1 - tan\delta')^3$
$tan\delta' = \frac{H}{V'}\ for\ c' = 0$	$tan\delta' = \frac{H/V'}{\left(1 + \frac{B'L'c'}{V'tan\varphi'}\right)}\ for\ c' > 0$	$B' = B - 2e_x;\ L' = L - 2e_y;\ e_x = \frac{d}{tan\alpha}$ 
Fine-grained soils (undrained clays; less accurate approximation if used for sands and gravels)		
$q_f = 5.14 \cdot C_u \cdot s_c \cdot i_c + p_{ov};\ s_c = 1 + 0.2 \cdot \left(\frac{B'}{L'}\right);\ i_c = 0.5 \cdot \left(1 + \left(\sqrt{1 - \frac{H'}{C_u A'}}\right)\right)$ where H' is the horizontal force and A' is the area: $A' = B' \cdot L'$ (rectangular) $A' = \pi \cdot (D'/2)^2$ (circular with D': diameter); base and ground factors ignored.		

Bearing capacity in piles:

$Q_u = Q_s + Q_b - W_p$	
Cohesionless soils:	Cohesive soils:
$Q_s = q_s \cdot A_s = A_s \cdot k_s \cdot \sigma_{v0} \cdot \tan \delta$	$Q_s = q_s \cdot A_s = \alpha \cdot C_u \cdot A_s$
$Q_b = q_b \cdot A_b = N_q \cdot \sigma'_{v0} \cdot A_b$	$Q_b = q_b \cdot A_b = N_c \cdot C_{ub} \cdot A_b$

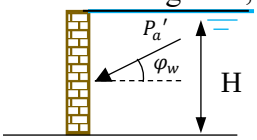
Passive and active pressures:

$\sigma'_h = K_a \cdot \sigma'_v$ (active)	$\sigma'_h = K_p \cdot \sigma'_v$ (passive)

continued

RANKINE'S theory: H high of the layer of soil behind the wall retaining wall, φ' is the friction angle and c' is the cohesion

In the following, $P_a = P_a' + U$ and $P_p = P_p' + U$,
where U is the resultant of the pore water pressure.



$Obs.\ \varphi_w = 0\ (Rankine's)$

$K_a = tan^2\left(\frac{\pi}{4} - \frac{\varphi'}{2}\right) = \left(\frac{cos\,\varphi'}{1 + sin\,\varphi'}\right)^2 = \frac{1 - sin\,\varphi'}{1 + sin\,\varphi'}$	$K_p = tan^2\left(\frac{\pi}{4} + \frac{\varphi'}{2}\right) = \left(\frac{cos\,\varphi'}{1 - sin\,\varphi'}\right)^2 = \frac{1 + sin\,\varphi'}{1 - sin\,\varphi'}$

Cohesionless soils:	Cohesive: (Undrained)	Cohesive: (drained)
$\sigma'_h = K_a \sigma'_v$	$\sigma_h = \sigma_v - 2C_u$	$\sigma_h' = \sigma_v' K_a - 2c'\sqrt{K_a}$
$P_a' = \frac{1}{2} K_a \gamma' H^2$	$P_a = \frac{1}{2}(\gamma H - 2C_u)\left(H - \frac{2C_u}{\gamma}\right)$	$P_a' = \frac{K_a \gamma'}{2}\left(H - \frac{2c'}{\gamma'\sqrt{K_a}}\right)^2$
$\sigma'_h = K_p \sigma'_v$	$\sigma_h = 2C_u + \sigma_v$	$\sigma_h' = \sigma_v' K_p + 2c'\sqrt{K_p}$
$P_p' = \frac{1}{2} K_p \gamma' H^2$	$P_p = \frac{H}{2}(\gamma H + 4C_u)$	$P_p' = \frac{1}{2} K_p \gamma' H^2 + 2c'\sqrt{K_p} H$

COULOMB'S theory: Friction and adhesion between soil and wall are φ_w and c_w, resp.

$K_a = \left(\frac{cos\,\varphi'}{1 + sin\,\varphi'\sqrt{1 + \frac{tan\,\varphi_w'}{tan\,\varphi'}}}\right)^2 \frac{1}{cos\,\varphi_w'}$	$K_p = \left(\frac{cos\,\varphi'}{1 - sin\,\varphi'\sqrt{1 + \frac{tan\,\varphi_w'}{tan\,\varphi'}}}\right)^2 \frac{1}{cos\,\varphi_w'}$

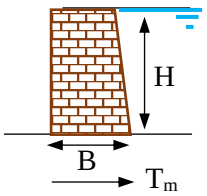
Cohesionless:	Cohesive: (Undrained)	Cohesive: (drained)
$\sigma'_h = K_a cos\varphi_w' \sigma'_v$	$\sigma_h = \sigma_v - 2C_u\sqrt{1 + \frac{C_w}{C_u}}$	$\sigma_h' = (K_{a1}\sigma_v' - c'K_{a2}) cos\varphi_w'$
$P_a' = \frac{1}{2} K_a \gamma' H^2$	$P_{a,h} = \frac{\gamma}{2}\left(H - \frac{2C_u}{\gamma}\sqrt{1 + \frac{C_w}{C_u}}\right)^2$	$P_a' = \frac{K_{a1}\gamma'}{2}\left(H - \frac{c'K_{a2}}{\gamma' K_{a1}}\right)^2$
$\sigma'_h = K_p cos\varphi_w \sigma'_v$	$\sigma_h = \gamma z + 2C_u\sqrt{1 + \frac{C_w}{C_u}}$	$\sigma_h' = \left(\gamma z K_{p1} + c'K_{p2}\right) cos\varphi_w'$
$P_p' = \frac{1}{2} K_p \gamma' H^2$	$P_{p,h} = \frac{H}{2}\left(\gamma H + 4C_u\sqrt{1 + \frac{C_w}{C_u}}\right)$	$P_p' = \frac{1}{2} K_{p1}\gamma' H^2 + c'K_{p2} H$

where, $K_{a1} = K_a$, $K_{p1} = K_p$ and

continued

$K_{a2} = \frac{\cos\varphi'\left(2\sqrt{1+\frac{c_w'}{c'}} + \left(2+\frac{c_w'}{c'}\right)\sin\varphi'\right)}{\left(1+\sin\varphi'\sqrt{1+\frac{\tan\varphi_w'}{\tan\varphi'}}\right)^2 \cos\varphi_w'}$	$K_{p2} = \frac{\cos\varphi'\left(2\sqrt{1+\frac{c_w'}{c'}} - \left(2+\frac{c_w'}{c'}\right)\sin\varphi'\right)}{\left(1-\sin\varphi'\sqrt{1+\frac{\tan\varphi_w'}{\tan\varphi'}}\right)^2 \cos\varphi_w'}$
Cantilever wall in sand:	$FoS = \frac{N'\tan\varphi_w'}{T_m} = \frac{(W-U)\tan\varphi_w'}{T_m}$
Gravity wall in sand:	$FoS = \frac{N\tan\varphi_w'}{T_m}$

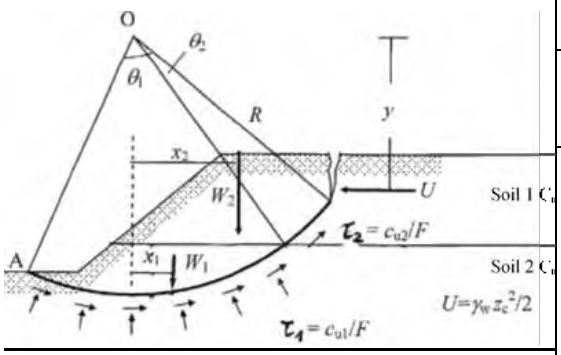
Gravity wall in clay:



(Undrained) $FoS = \frac{C_w B}{T_m}$

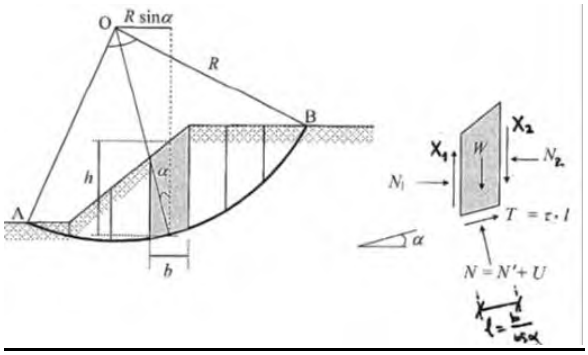
(Drained) $FoS = \frac{N'\tan\varphi_w' + c_w' B}{T_m}$

Slope stability: (Note: for undrained strength in infinite slope take c'=C_u and φ'=0)

Infinite slope with parallel flow: H: layer thickness, m: position of water table from bedrock, β: slope angle; γ_1 (dry soil; m=0), γ_2 (saturated soil; m=1).	$FoS = \frac{c' + H\left(\gamma_1(1-m) + m(\gamma_2-\gamma_w)\right)\cos^2\beta\tan\varphi'}{H\left((1-m)\gamma_1 + m\gamma_2\right)\cos\beta\sin\beta}$
Vertical cut (dry soil):	$FoS = \frac{4C_u}{H\gamma\sin 2\alpha}$ (undrained) $FoS = \frac{4(c'+\sigma'\tan\varphi')}{H\gamma\sin 2\alpha}$ (drained)
Circular slope failure: (undrained) 	$FoS = \frac{R^2\theta C_u}{Wx}$ $FoS = \frac{R^2\theta C_u}{Wx+Uy}$ (with cracks) $FoS = \frac{(\theta_1 C_{u1} + \theta_2 C_{u2})R^2}{W_1x_1 + W_2x_2 + Uy}$ (layered soil)

$\sin 2\alpha = 2\sin\alpha\cos\alpha$, $\cos 2\alpha = \cos^2\alpha - \sin^2\alpha$ and arc-length: $\widehat{AB}$ = Rθ (θ is in radians).

continued

Circular: (drained):  	$$FoS = \frac{\sum_{i=1}^{n} l(c' + \sigma' tan\,\varphi')_i}{\sum_{i=1}^{n}(W\,sin\,\alpha)_i} = \frac{\sum_{i=1}^{n}(lc' + N' tan\,\varphi')_i}{\sum_{i=1}^{n}(W\,sin\,\alpha)_i}$$
Modified Bishop's b: width of the slice, l: length of the slice, α: inclination of the slice to the horizontal direction.	$$FoS = \frac{\sum_{i=1}^{n}(bc' + (W - U\,cos\,\alpha)\,tan\,\varphi')_i}{\sum_{i=1}^{n}(W\,sin\,\alpha)_i} M(\alpha)$$ where, $$M(\alpha) = \frac{1}{cos\,\alpha\left[1 + \frac{tan\,\alpha\,tan\,\varphi'}{FoS}\right]}$$
Fellenius:	$$FoS = \frac{\sum_{i=1}^{n}(lc' + (W\,cos\,\alpha - ul)\,tan\,\varphi')_i}{\sum_{i=1}^{n}(W\,sin\,\alpha)_i}$$

END